

$\mu_{\bar{X}} = \sum \bar{x} * f(\bar{x})$ $\sigma_{\bar{X}}^2 = \sum (\bar{x} - \mu_{\bar{X}})^2 * f(\bar{x})$	$\mu_{\bar{X}} = \mu$ $\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$ $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$
$\mu_{\bar{x}_1 - \bar{x}_2} = \sum (\bar{x}_1 - \bar{x}_2) * f(\bar{x}_1 - \bar{x}_2)$ $\sigma_{\bar{x}_1 - \bar{x}_2}^2 = \sum \left((\bar{x}_1 - \bar{x}_2) - \mu_{\bar{x}_1 - \bar{x}_2} \right)^2 * f(\bar{x}_1 - \bar{x}_2)$	$\mu_{\bar{x}_1 - \bar{x}_2} = \mu_1 - \mu_2$ $\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
$\hat{P} = \frac{X}{n}$ $\mu_P = P$ $\sigma_P = \sqrt{\frac{P(1-P)}{n}}$	$\mu = \frac{\sum_{i=1}^N x_i}{N}$ $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$ $\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$ $S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$
$\bar{X} - Z_{\frac{\alpha}{2}} * \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + Z_{\frac{\alpha}{2}} * \frac{\sigma}{\sqrt{n}}$ $\bar{X} - Z_{\frac{\alpha}{2}} * \frac{s}{\sqrt{n}} < \mu < \bar{X} + Z_{\frac{\alpha}{2}} * \frac{s}{\sqrt{n}}$ $\bar{X} - t_{\frac{\alpha}{2}} * \frac{s}{\sqrt{n}} < \mu < \bar{X} + t_{\frac{\alpha}{2}} * \frac{s}{\sqrt{n}}$	$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$
$(\bar{x}_1 - \bar{x}_2) - Z_{\frac{\alpha}{2}} * \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + Z_{\frac{\alpha}{2}} * \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ $(\bar{x}_1 - \bar{x}_2) - Z_{\frac{\alpha}{2}} * \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + Z_{\frac{\alpha}{2}} * \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$
$(\bar{x}_1 - \bar{x}_2) - t_{\frac{\alpha}{2}} * s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2$ $< (\bar{x}_1 - \bar{x}_2) + t_{\frac{\alpha}{2}} * s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ Where $s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}}$ and $df = n_1 + n_2 - 2$	: $(\bar{x}_1 - \bar{x}_2) - t_{\frac{\alpha}{2}} * s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2$ $< (\bar{x}_1 - \bar{x}_2) + t_{\frac{\alpha}{2}} * s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ Where: $s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}}$ and $df = n_1 + n_2 - 2$
: $(\bar{x}_1 - \bar{x}_2) - t_{\frac{\alpha}{2}} * s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2$ $< (\bar{x}_1 - \bar{x}_2) + t_{\frac{\alpha}{2}} * s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ Where $s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}} \text{ and } df = n_1 + n_2 - 2$	: $\frac{(n-1)s^2}{\chi_{\frac{\alpha}{2}}^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_{1-\frac{\alpha}{2}}^2}$ Where $df=n-1$ $s^2 = \frac{n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2}{n(n-1)}$

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$(\bar{x}_1 - \bar{x}_2) - t_{\frac{\alpha}{2}} * \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\frac{\alpha}{2}} * \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$ Where $df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\left[\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1}\right] + \left[\frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}\right]}$	$\frac{s_1^2}{s_2^2} \times \frac{1}{f_{\frac{\alpha}{2}}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2} f_{\frac{\alpha}{2}}(v_2, v_1)$ Where $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$
Where $\bar{d} - t_{\frac{\alpha}{2}} * \frac{s_d}{\sqrt{n}} < \mu_d < \bar{d} + t_{\frac{\alpha}{2}} * \frac{s_d}{\sqrt{n}}$ $s_d = \sqrt{\frac{n \sum d_i^2 - (\sum d_i)^2}{n(n-1)}}$	$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$ $t = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$
$\hat{p} - z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} < p < \hat{p} + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ $n\hat{p} \geq 5$ and $n(1-\hat{p}) \geq 5$ $(\hat{p}_1 - \hat{p}_2) - z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}} < p_1 - p_2$ $< (\hat{p}_1 - \hat{p}_2) + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$ If $n_1 \hat{p}_1 \geq 5, n_1 \hat{q}_1 \geq 5, n_2 \hat{p}_2 \geq 5, \text{ and } n_2 \hat{q}_2 \geq 5$	$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ $z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ $t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ $s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$ $df = n_1 + n_2 - 2$
$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\left[\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1}\right] + \left[\frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}\right]}$ $t = \frac{\bar{d} - d_0}{\frac{s_d}{\sqrt{n}}}$ $s_d = \sqrt{\frac{n \sum d_i^2 - (\sum d_i)^2}{n(n-1)}}$	$z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$ $Z = \frac{(\hat{P}_1 - \hat{P}_2) - (p_1 - p_2)}{\sqrt{p_1 q_1 / n_1 + p_2 q_2 / n_2}}$